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Research on the balancing problem of human–robot collaborative assembly line in SMEs considering ergonomic risk and cost

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Research on the balancing problem of human-robot collaborative assembly line
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Abstract

As the development of Industry 5.0, an increasing number of collaborative robots (cobots) are integrated into manufacturing systems to assist workers. The inappropriate allocation of tasks and resources (workers and cobots) in human-robot collaborative assembly lines will result in problems such as production efficiency decrease. Due to the lack of research on assembly line balancing problem with human-robot collaboration (ALBP-HRC) for small and medium-sized enterprises (SMEs), this study develops a novel multi-objective optimization mathematical model based on the cost-conscious characteristics of SMEs. The model aims to simultaneously minimize cycle time, ergonomic risk, and cost. To solve the model, a problem-specific improved multi-objective particle swarm optimization algorithm (IMOPSO) with a specialized initialization scheme and efficient local search method is developed, and a novel three-vector encoding and decoding scheme is proposed. Subsequently, the computational tests demonstrate that the IMOPSO algorithm outperforms the comparison algorithms in most results (accounting for 87.7%), while exhibiting superior average performance across all test cases. Finally, human-robot collaborative balancing schemes for a real electric vehicle air conditioner assembly line are designed, providing a feasible solution for the assembly line balancing problem of similar electromechanical products in SMEs.

Keywords: Human-robot collaboration, Assembly line balancing, Particle swarm algorithm, Multi-objective optimization, Cost, Ergonomic risk

1. Introduction

Small and medium-sized enterprises (SMEs) are the main contributors in most industries and countries [1], and their development and future are an inalienable part of Industry 5.0. Assembly lines are widely used in SMEs, in which the final assembly of the product from previously manufactured parts is performed [2]. Repetitive or strenuous tasks on the assembly line often result in worker fatigue or injury. For example,

musculoskeletal disorders common on the assembly line cause approximately 43% of the total annual lost-time [3]. Occupational diseases are more prevalent in SMEs with a lower degree of automation and fewer preventive measures, and traditional assembly line balancing methods struggle to reduce ergonomic risk effectively. Meanwhile, SMEs usually adopt a customized production mode, which requires the highly flexible assembly line. Fully-automated assembly lines composed of traditional industrial robots are both expensive and lack flexibility. SMEs should focus on upgrading the most critical machines and thus limit investment costs [4]. Achieving semi-automation of the assembly line at an appropriate cost is more desired by SMEs. Thus, collaborative robots (cobots), as industrial robots specifically designed to interact with human workers with high flexibility [5], are introduced into assembly lines. Cobots reduce human ergonomic risk [6] and increase production efficiency [7] by assisting in tasks, which aligns with the human-centric features of Industry 5.0 manufacturing. The human-robot collaboration assembly line has three operation modes, which are human operation, cobots operation and human-robot collaboration. The three operation modes are shown in Figure 1. In Workstation 1, a worker and a cobot collaborate on assembly tasks. In workstation 2 and workstation 3, a worker and a cobot independently operate the assembly tasks, respectively. Due to the presence of various operation modes on the assembly line, it is difficult to achieve optimal goals of production such as reducing cycle time, without well-designed task and resource allocation schemes in workstations. This has prompted research on the assembly line balancing problem with human-robot collaboration (ALBP-HRC).

In ALBP-HRC, due to the flexibility and capability limitations of cobots, some tasks cannot be operated by cobots independently. Similarly, some tasks cannot be operated collaboratively by workers and robots [6] because these tasks are not suitable for collaboration, and performing them in collaboration mode would actually reduce efficiency. Another important constraint is the precedence constraint, which limit the sequence of task allocation. These two constraints are systematically described in graphical form. Figure 2 is a small case of a precedence relationship diagram that describes the constraints in an assembly line, where each number represents a specific assembly task. If an arrow points from task A to task B, it means that task A is an immediate predecessor of task B. Task B may have multiple immediate predecessors and can only be operated after all of its immediate predecessors are completed. Different identification colors denote various

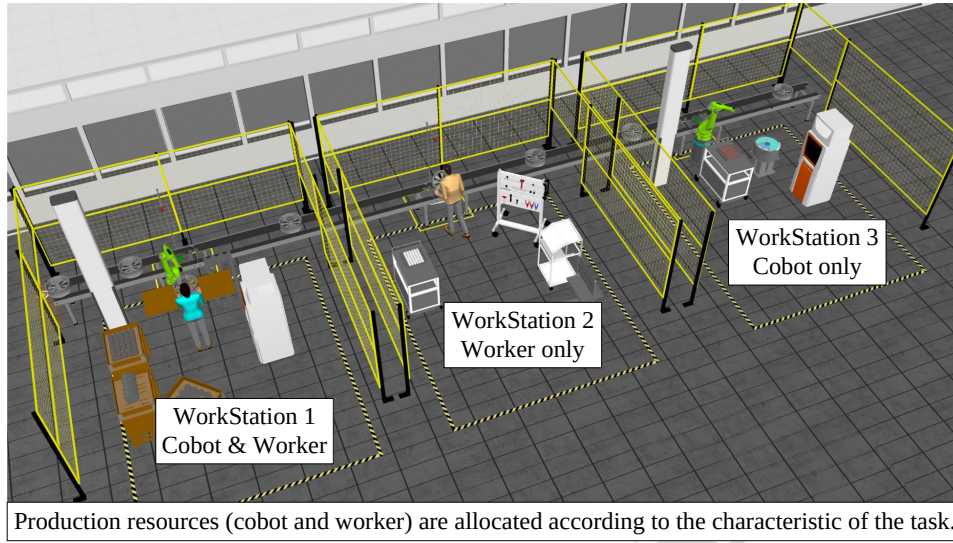


Figure 1: An illustration of the human-robot collaborative assembly line with three different modes of operation

operation constraints. All the constraints make the ALBP-HRC a complex optimization problem.

For SMEs, the cost changes caused by introducing cobots into the assembly line cannot be ignored. According to the reviews by Chutima [8] and Kheirabadi [9], there is a lack of research and literature on human-robot collaborative assembly line balancing in SMEs that targets the simultaneous minimization of cycle time, ergonomic risk and cost, simultaneously. Therefore, this study aims to develop an approach to solve the valuable three-objective ALBP-HRC. The main contributions of this study are as follows:

1) Considering the characteristics of assembly lines in SMEs, a novel multi-objective ALBP-HRC is studied, and a targeted investigation is conducted on an existing electric vehicle air conditioner assembly line.

2) A novel mixed-integer programming (MIP) model is proposed that is specifically oriented to the characteristics of the multi-objective optimization problem.

3) A novel three-vector encoding scheme and an improved multi-objective particle swarm optimization algorithm (IMOPSO) are proposed. The performance of IMOPSO in solving the ALBP-HRC problem has been validated through comparisons with various multi-objective optimization algorithms.

The subsequent sections are organized as follows: In section 2, the literature review is provided. In section 3, the characteristics of human-robot collaborative assembly line problems in SMEs are described,

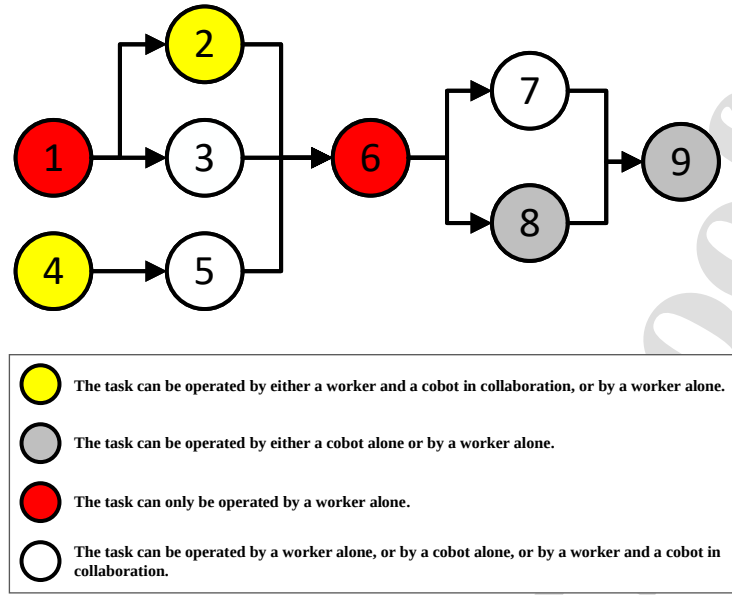


Figure 2: A case of precedence relationship diagram

and a mathematical model is constructed. In section 4, the proposed IMOPSO algorithm is developed. In section 5, the comparative experiments between the IMOPSO and other algorithms are presented. In section 6, the case application of an electric vehicle air conditioner assembly line is analyzed. Finally, section 7 presents conclusion and future research.

2. Literature review

Since Salveson published the first research paper on the assembly line balancing problems in 1955, assembly line balancing problems have become a research hotspot of mutual concern to researchers [10]. The assembly line balancing problem has been proven to be an NP-hard problem [11]. Therefore, the more complex ALBP-HRC is also an NP-hard problem. Since this study focuses on modeling the human-robot collaborative assembly line balancing problem and the optimization algorithm, the research status of the ALBP-HRC and the particle swarm optimization algorithm (PSO) are reviewed in detail. Additionally, Industry 5.0 and human-robot collaboration are also reviewed to provide a stronger research background.

2.1. Assembly line balancing problem with human-robot collaboration

Since 2019, research papers focusing on ALBP-HRC have gradually emerged. Samouei and Ashayeri [12] investigated two models for semi-automated mixed-model assembly line balancing, and the study used the Cplex solver to solve the experiments. Dalle Mura and Dini [13] employed a genetic algorithm (GA) to solve the straight ALBP-HRC, considering workers and robots with varying abilities while ensuring they possess the minimum skills required for task performance. In contrast to other studies, they assumed that robots exhibit faster task completion times compared to workers. Their study integrates the three objectives of minimizing the assembly cost, minimizing the number of skilled workers, and balancing worker energy load into a unified objective function. Weckenborg et al. [14] developed an MIP model for ALBP-HRC, aiming to reduce cycle time under the condition of fixed workstations, with options of one type of worker and one type of robot resource available. A hybrid GA is developed to solve this NP-Hard problem. Koltai et al. [15] established three models for optimizing the number of workers, robots, and cycle time in the straight ALBP-HRC. They also proposed the assumption that tasks assigned to the same group should not be more than one workstation apart, and experiments are conducted using the CPLEX solver on a power inverter assembly line with 49 assembly tasks. Nourmohammad et al. [7] developed an MIP model to optimize the cycle time and the total number of employees and cobots, and designed a neighborhood-search simulated annealing algorithm to solve the model. Stecké and Mokhtarzadeh [16] investigated the straight mixed-model ALBP-HRC considering the situation where cobots can move and employed the Benders decomposition algorithm to address the problem. Li et al. [17] conducted research on the single-objective U-shaped ALBP-HRC and explored multiple types of cobots under budget constraints, aiming to reduce cycle time in a fixed workstation scenario. To tackle large-size instances, they developed the improved artificial bee colony algorithm and the migrating bird optimization algorithm. Zacharia et al. [18] studied a fuzzy ALBP-HRC aimed at minimizing both the fuzzy cycle time and the fuzzy smoothness index. They proposed a problem-specific GA as the solution method, which incorporates a specific encoding scheme and evaluation mechanism. Huang et al. [6] investigated the human-robot collaborative mixed-model two-sided assembly line balancing problem and proposed a highly effective multi-objective discrete artificial bee colony algorithm with specialist bees. Considering the background and mathematical model of Huang's research, it can be regarded as being oriented towards large enterprises. Mao

et al. [19] studied the parallel human–robot collaboration assembly line worker assignment and balancing problem, developed a MIP model to minimize cycle time and proposed tight lower and upper bounds. To handle the model, the study developed an adapted tabu search algorithm. Cai et al. [20] studied the ALBP-HRC considering assembly complexity and workload balance. They proposed an improved multi-objective migratory bird optimization algorithm with fast non-dominated sorting and demonstrated its effectiveness through computational experiments.

In summary, most studies on ALBP-HRC have formulated corresponding MIP models and employed heuristic algorithms for solving them, but few have specifically discussed the problem context and the type of enterprise. The main literature on the ALBP-HRC is summarized in Table 1.

Table 1: Summary of literature on the ALBP-HRC

Paper (sorted by years)	Special case application	Type of enterprise	Optimization criteria	Solution method
Samouei and Ashayeri [12]	-	Unspecified	Min cost and cycle time	CPLEX
Dalle Mura and Dini [13]	Scooter chassis	Unspecified	Min assembly line cost, the number of skilled workers, the energy load variance among workers	GA
Weckenborg et al. [14]	-	Unspecified	Min cycle time	GA
Koltai et al. [15]	Power inverters	Unspecified	Min stations and cycle time	CPLEX
Nourmohammadi et al. [7]	Mass balancing system	Unspecified	Min cycle time and number of worker	Simulated Annealing
Stecke and Mokhtarzadeh [16]	Balance shaft module	Unspecified	Min cycle time and ergonomic risk	Benders decomposition Algorithm
Li et al. [17]	-	Unspecified	Min cycle time	Artificial bee colony algorithm and migrating bird optimization algorithm
Zacharia et al. [18]	-	Unspecified	Min fuzzy cycle time and fuzzy smoothness index	GA
Huang et al. [6]	Automobile engine	Large enterprises	Min energy consumption, ergonomic risk, cycle time	Artificial bee colony algorithm
Mao et al. [19]	-	Unspecified	Min cycle time	Tabu search algorithm
Cai et al. [20]	-	Unspecified	Min cycle time, assembly complexity and workload balance index	Migratory bird optimization algorithm
Current study	Electric vehicle air conditioner	SMEs	Min cycle time, ergonomic risk and cost	IMOPSO

2.2. Particle swarm algorithm for assembly line balancing

Heuristic algorithms are often used to solve complex assembly line balancing problems and can yield good results. Different algorithms exhibit varying degrees of competitiveness on different types of problems. As a classic algorithm, the PSO algorithm possesses strong versatility, making it worthwhile to improve it for the ALBP-HRC. PSO is first proposed by Kennedy and Eberhart [21], and subsequently, the multi-objective particle swarm optimization algorithm (MOPSO) is developed by Colello et al. [22], and demonstrated that MOPSO exhibits strong competitiveness in terms of performance for multi-objective optimization problems.

Hamta et al. [23] proposed a method based on the combination of PSO with variable neighborhood search to solve the multi-objective optimization of a single-model assembly line balancing problem. Through comparison with various multi-objective evolutionary algorithms in testing problems, the superior efficiency of PSO/VNS has been demonstrated. Mukund Nilakantan et al. [24] proposed the use of PSO and CS-PSO algorithms to solve the problem of robotic assembly line balancing. In cases where the optimization objective is to minimize cycle time, the performance of these algorithms is generally better than that of IBM CPLEX Optimization solver. Masoud Rabbani et al. [25] proposed a MOPSO algorithm to solve the problem of U-shaped robotic assembly line balancing. The mathematical model is established with two types of tasks, aiming to minimize robot purchasing costs, robot setup costs, sequence-dependent setup costs, and cycle time. Su et al. [26] proposed a PSO with a various population strategy to optimize the hybrid assembly sequence planning and assembly line balancing problems. The MOPSO algorithm have potential in solving the ALBP-HRC.

2.3. Industry 5.0 and human-robot collaboration

As a new generation of production models, Industry 5.0 emphasizes placing humans at the key of the production system, collaborating with efficient and intelligent machines to achieve better manufacturing goals [27]. In the research on cobots within the context of Industry 4.0 [28], the primary concern lies in their contributions to enterprise automation and production efficiency. With the continuous intensification of research on the control of key robotic manipulators [29, 30] and the progressive combination of human-robot collaboration and artificial intelligence [31], it has vigorously bolstered confidence in the broader application of cobots in enterprises.

Maddikunta et al. [32] investigated the enabling technologies and potential applications of Industry 5.0, highlighting that cobots not only help improve productivity and safety but also enhance a company's competitiveness in a highly competitive market environment. Ivanov [33] proposed that Industry 5.0 is a combination of organizational principles and technologies to design and manage operations and supply chains as resilient, sustainable, and human-centric systems, emphasizing the importance of multidisciplinary technologies such as automation and human-robot collaboration. Rožanec et al. [34] presented an architecture that integrates Artificial Intelligence, simulated reality, decision-making, and users' feedback, focusing on synergies between humans and machines, thereby supporting the application of digital twins and human-robot collaboration in Industry 5.0. Coronado et al. [35] surveyed and classified performance metrics and measures for Human-Robot Interaction systems, providing an in-depth discussion of factors related to human-robot collaboration, thereby contributing to the progression of Industry 5.0. It is foreseeable that cobots are emerging as a key force in realizing human-centered manufacturing [36].

2.4. Research gap

The literature review reveals that research on ALBP-HRC has become increasingly advanced, yet the scope and solution methods still exhibit certain limitations. As a new research direction, early studies on ALBP-HRC primarily employed GA and CPLEX as popular solution methods. While general-purpose approaches have demonstrated some effectiveness in problem-solving, few studies have explored how algorithms can be adapted to the specific characteristics of ALBP-HRC to achieve better efficiency and accuracy. Whereas previous studies have demonstrated that PSO is effective in solving assembly line balancing problems, its application to ALBP-HRC remains largely unexplored. Although research on multi-objective optimization of ALBP-HRC account for the majority, some research integrates multiple objectives into a single function, focusing more on theoretical exploration while lacking practical discussions. Meanwhile, due to the lack of real-world case applications, research on ALBP-HRC has predominantly relied on simulation. Analyzing common electromechanical product assembly lines could provide valuable insights for addressing real-world challenges. Most studies on ALBP-HRC have primarily focused on improving production efficiency, with limited discussion on the practical needs of different types of enterprises. Currently, there is a lack of multi-objective optimization research on ALBP-HRC in SMEs, particularly that aims to minimize cycle time,

ergonomic risk, and cost. Therefore, this study constructs a novel multi-objective ALBP-HRC model and proposes the IMOPSO algorithm, aiming to fill the gaps in existing research and contribute to the in-depth study of ALBP-HRC in SMEs.

3. Problem description and mathematical model formulation

3.1. Problem description

In the problem scenario of this paper, the assembly line cannot achieve full automation due to the limited capabilities of cobots. Distinct from the assembly lines of large electromechanical products, the tasks on the assembly lines within SMEs generally do not demand sophisticated skills. Considering that workers can operate tasks within the specified time as per the Standard Operating Procedure after simple training, it is assumed that workers have similar capabilities and possess the ability to complete all tasks.

The patterns of collaboration on different human-robot collaborative assembly lines may be different. According to Hashemi-Petroodi et al. [37] and Li et al. [17], there are four types of human-robot collaboration within the same workstation on assembly lines, as shown in Table 2. In light of the restricted size and volume of the principal products in SMEs, parallel operations might cause collisions. Consequently, this paper focuses on investigating the third type, where each workstation can be assigned at most one worker and one cobot. A worker and a cobots cannot operate different tasks in parallel. They can either sequentially operate tasks or collaborate to operate common tasks.

The first optimization objective of ALBP-HRC is to reduce the cycle time. The second optimization objective is to reduce ergonomic risk. With the economic development of society, the government and society generally advocate increasing the protection of workers. The manual completion of tasks by workers will consume physical energy, and excessive workload can pose potential ergonomic risks. Full automation of labor through cobots would eliminate the workload on workers entirely, while human-robot collaboration in operating tasks would alleviate the workload on workers. Many scholars who study the balance of assembly line problems have proposed various methods to measure ergonomic risk, mainly based on the assessment of physical energy consumption, such as Dalle Mura and Dini [38], Stecke and Mokhtarzadeh [16], Weckenborg et al. [39]. Considering the wide variety of product types in SMEs, a more universal ergonomic risk assessment

Table 2: Four types of human-robot collaboration

Number of types	Way of cooperation	Way of operation	Description
1	Separate cooperation	Sequential operation	Without common tasks, tasks can only be operated sequentially
2	Separate cooperation	Simultaneous operation	Without common tasks, tasks can be operated in parallel
3	Collaborative cooperation	Sequential operation	With common tasks, tasks can only be operated sequentially
4	Collaborative cooperation	Simultaneous operation	With common tasks, tasks can be operated in parallel

method is appropriate. The energy consumption method proposed by Stecke and Mokhtarzadeh is shown as Equation 1, which is used as part of a measure of ergonomic risk. For example, according to Garg's proposed formula [40], an adult male weighing 70 kg who lifts a 10 kg load with his arm from 1 meter to 1.5 meters will consume approximately 0.164 kcal. For the same worker, maintaining a standing posture consumes 1.68 kcal per minute. Assuming that the operation time of this task is 5 seconds, this worker would consume 0.248 kcal.

$$EE = \sum_{a=1}^A P_a u_a + \sum_{b=1}^B W_b v_b \quad (1)$$

Where, EE is the energy expenditure, A is the number of body postures, B is the number of basic movements, P_a is the basal energy metabolic rate for maintaining a posture, u_a is the duration of posture a , W_b is the basal energy metabolic rate for performing basic motion, and v_b is the duration of performing motion b .

The third objective of optimization is to control cost. Enterprises consider return on investment (ROI) as a key factor when making investments, especially for SMEs with limited funds. The introduction of cobots requires careful cost control. Incorporating cobots into the manual assembly line changes the number of workers and cobots, thus generating a change in cost. More expensive cobots have greater flexibility and capabilities, enabling them to complete assembly tasks more quickly [41], which presents a trade-off between cost and cycle time. The cost of assembly line workers is primarily the annual salary. The cost of introducing

cobots is primarily the high purchase price [42], followed by ancillary costs such as installation, accessories, and maintenance[43]. To obtain a reasonable estimate of the annual comprehensive cost of using cobots, the annuity method is employed [44]. The formula for the annuity method is shown in Equation 2. For example, consider a general cobot with a purchase price of 30,000 euros, assuming the total investment, including ancillary costs, is 48,000 euros. With a uniform discount rate of 0.07 and an economic life of the investment of 5 years [45], the annual comprehensive cost of this cobot would be 11,707 euros.

$$AC = TI \times \frac{(1 + I)^E \times I}{(1 + I)^E - 1} \quad (2)$$

Where, AC is the annual comprehensive cost of the cobot, TI is the total investment of the cobot, I is the uniform discount rate, E is the economic life of the investment.

3.2. Assumptions of the model

To establish the MIP model, based on the information in the problem description, the following assumptions are clarified in this study. Assumption 1 to Assumption 5 are typical assumptions for Type-II ALBP-HRC. Assumption 6 and Assumption 7 conform to the problem description. Assumption 8 is because of the fact that the resources and time that SMEs can mobilize are usually not abundant. Assumptions 9 to 11 are assumptions with sources and practical application values.

- 1) One assembly line assembles only one type of product.
- 2) The number of workstations is known and fixed, as with other type-II assembly line problems.
- 3) One workstation can be assigned a worker, a cobot, or a worker and a cobot combination.
- 4) The precedence relationships between assembly tasks are known.
- 5) The operation constraints for each assembly task are known.
- 6) The workers are regarded as identical, with only one type and the ability to complete any assembly task.
- 7) In the same workstation, the worker and the cobot can only operate a common task at a moment.
- 8) For an assembly line, the types of cobots that can be introduced are known and limited. Different types of cobots have varying capabilities and costs.

9) The operation time for completing tasks varies when using different combinations of cobots and workers [17].

10) The energy consumption of workers varies when collaborating with different cobots [16].

11) According to the principle of maximizing the utilization of cobots and considering the tireless characteristic of cobots [46], if a workstation both has a worker and a cobot, tasks that can be operated in collaboration mode will not be assigned solely to the worker.

3.3. Notations

Notation	Definition
Indices and Parameters:	
NT	Total number of tasks
ST	Set of tasks, $ST = \{1, 2, 3, \dots, NT\}$
i, j	Task index, $i, j \in \{1, 2, 3, \dots, NT\}$
NW	Total number of workstations
SW	Set of workstations, $SW = \{1, 2, 3, \dots, NW\}$
k	Workstation index, $k \in \{1, 2, 3, \dots, NW\}$
NC	Total number of cobots types
c	Cobots type index, $c \in \{1, 2, 3, \dots, NC\}$
v	Process method index, $v \in \{1, 2, 3, \dots, 2NC+1\}$. When $v=1$, the process method for the task is operated by a worker; when $2 \leq v \leq NC+1$, the process method for the task is operated by a specific type of cobot; when $NC+2 \leq v \leq 2NC+1$, the process method for the task is operated by collaboration between a specific type of cobot and a human worker

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(Continued)

t_{iv}	Operation time for task i by process method v . If task i cannot be operated by process v , t_{iv} is null
e_{iv}	Energy consumption (kcal) of a worker when the task i is operated by process method v
TS_i	Start time of task i
AC_c	Annual comprehensive cost of c type of cobot
AS	Annual salary of workers
$P(j)$	Set of immediate predecessors of task j
M	A large positive number, set to 100,000 usually
Decision variables:	
CT	Cycle time
$COST_{w\&c}$	Annual comprehensive cost for workers and cobots in the assembly line
ERI	Ergonomic risk index
P_{ij}	equals to 1, if task i is the immediate successor of task j ; equals to 0, otherwise
A_{ik}	equals to 1, if task i is assigned to workstation k ; equals to 0, otherwise
Z_{iv}	equals to 1, if task i is operated by process method v ; equals to 0, otherwise
H_{ck}	equals to 1, if c type of cobot is assigned to workstation k ; equals to 0, otherwise
h_k	equals to 1, if a worker is assigned to workstation k ; equals to 0, otherwise

3.4. Mathematical model

Objective functions:

$$\text{Min } CT = \sum_{i=1}^{NT} \sum_{k=1}^{NW} \sum_{v=1}^{2NC+1} Z_{iv} t_{iv} A_{ik} \quad (3)$$

$$\text{Min } COST_{w\&c} = \sum_{k=1}^{NW} \sum_{c=1}^{NC} H_{ck} AC_c + \sum_{k=1}^{NW} h_k AS \quad (4)$$

$$\text{Min } ERI = \frac{\sum_{v=1}^{2NC+1} \sum_{i=1}^{NT} Z_{iv} e_{iv}}{\sum_{i=1}^{NT} e_{1v}} \quad (5)$$

Constraints:

$$\sum_{k=1}^{NW} A_{ik} = 1; \forall i \in ST \quad (6)$$

$$\sum_{v=1}^{2NC+1} \sum_{i=1}^{NT} A_{ik} Z_{iv} t_{iv} \leq CT; \forall k \in SW \quad (7)$$

$$\sum_{i=1}^{NT} A_{ik} \geq 1; \forall k \in SW \quad (8)$$

$$\sum_{k=1}^{NW} A_{jk} k - \sum_{k=1}^{NW} A_{ik} k \leq 0; i \in P(j) \quad (9)$$

$$TS_j - TS_i - \sum_{v=1}^{2NC+1} Z_{iv} t_{iv} \geq 0; k \in NW, i \in P(j), A_{ik} = A_{jk} = 1 \quad (10)$$

$$\sum_{v=1}^{2NC+1} Z_{iv} = 1; \forall i \in NT \quad (11)$$

$$\sum_{c=1}^{NC} H_{ck} \geq \sum_{v=2}^{2NC+1} Z_{iv} - M(1 - A_{ik}); k \in NW, i \in NT \quad (12)$$

$$h_k \geq Z_{1v} + \sum_{v=NC+2}^{2NC+1} Z_{iv} - M(1 - A_{ik}); k \in NW, i \in NT \quad (13)$$

$$\sum_{c=1}^{NC} H_{ck} \leq 1; \forall k \in NW \quad (14)$$

$$1 \leq h_k + \sum_{c=1}^{NC} H_{ck} \leq 2; \forall k \in NW \quad (15)$$

Equation 3 represents the objective function for minimizing cycle time. Equation 4 represents the objective function for minimizing cost, which is solely influenced by changes in resources. Equation 5 represents the objective function for the normalized index of minimizing ergonomic risk.

For constraints, Equation 6 ensures that each task is assigned to a workstation. Equation 7 ensures that the sum of processing times for tasks at any workstation does not exceed the assembly line cycle time, while also ensuring that the task within a workstation is not performed by incompatible process. Equation 8 ensures that each workstation is assigned at least one task. Equation 9 ensures that if task i is the immediate predecessor of task j , task i should be assigned to a workstation before or at the same workstation as task j . Equation 10 ensures that if task i is the immediate predecessor of task j and both tasks are on the same workstation, task j can only start after task i is completed. Equation 11 ensures that the process method of a task at a certain moment is fixed. Equation 12 ensures at least a cobot is assigned to the workstation, if there is a task in a workstation that is operated either by a cobot alone or collaboratively by a worker and a cobot. Equation 13 ensures at least a worker is assigned to the workstation, if there is a task in a workstation that is operated either by a worker alone or collaboratively by a worker and a cobot. Equation 14 ensures that the

number of cobots assigned to a workstation cannot exceed one. Equation 15 ensures that each workstation should be assigned either one worker or one cobot, with the constraint that only one worker and one cobot can be allocated at most.

4. Improved multi-objective particle swarm optimization algorithm for ALBP-HRC

MOPSO is a population-based evolutionary algorithm designed to find optimal solutions in complex search spaces, where each particle in the algorithm represents a solution. Due to its high efficiency, few parameters, and easy implementation, MOPSO has been widely used to solve NP problems, including assembly line balancing problems, as evidenced by the literature review. In preliminary tests, this study found that MOPSO demonstrated strong competitiveness under this model. Therefore, based on MOPSO and incorporating the characteristics of the ALBP-HRC, this study developed the IMOPSO algorithm. An analysis of existing ALBP-HRC research revealed a common issue: due to the presence of multiple alternative resources, task operation times often fluctuate within a large range, making it difficult to determine the initial cycle time for decoding. To address this, a novel initial cycle time update scheme is designed to reduce redundant calculations. Previous studies on multi-objective optimization algorithms mainly focused on enhancing the expansion of the search space and the exploration capability to escape local optima [6]. While such algorithms possess highly versatility, they lack discussions on incorporating problem-specific characteristics. In this study, a local search scheme is designed for IMOPSO, by considering bottleneck workstations in ALBP-HRC to effectively improve computational efficiency. Additionally, to enhance the performance of the initial solutions, a particle initialization strategy based on rule methods and mapping is developed. The main procedure of IMOPSO is illustrated in Figure 3, with key improvements highlighted in red.

In the IMOPSO algorithm, the methods for updating particle positions and velocities are similar to those in the traditional PSO algorithm. The position update formula is shown in Equation 16, and the velocity update formula is shown in Equation 17.

$$x_z(t+1) = x_z(t) + v_z(t+1) \quad (16)$$

Where $x_z(t+1)$ means the position of the particle in the solution space at time $t+1$, $v_z(t+1)$ means the

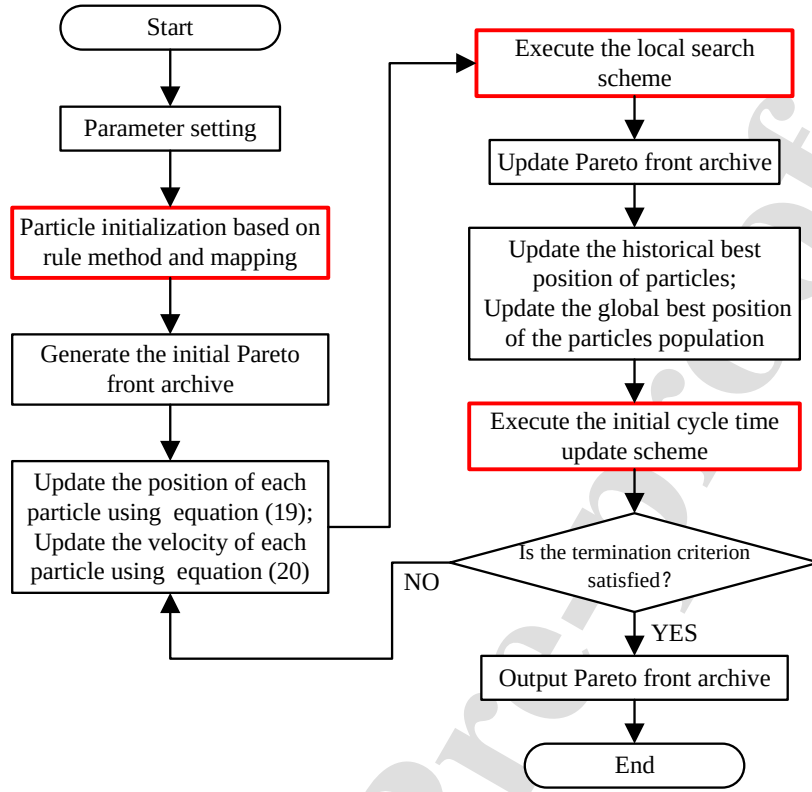


Figure 3: IMOPSO procedure

297 velocity of the z particle in the solution space at time $t + 1$.

$$v_z(t + 1) = \omega \times v_z(t) + c_1 r_1 \times (p_z^{best}(t) - x_z(t)) + c_2 r_2 \times (g^{best}(t) - x_z(t)) \quad (17)$$

298 Where ω represents the inertia factor, c_1 and c_2 represents the learning factor, and three parameters are set
 299 during the parameter tuning. $p_z^{best}(t)$ represents the local best position of the z particle in at t time, $g^{best}(t)$
 300 represents the global best position of all particles at t time, r_1 and r_2 are random numbers taken from the
 301 interval $[0,1]$.

302 In each iteration, the particle's current position is compared with its historical best position. If the current
 303 position is better, it replaces the particle's historical best position. In each iteration, a particle is selected
 304 from the current Pareto front solution set as the global best particle.

305 When dealing with discrete problems, the discretized PSO algorithm often achieves the same or even
 306 better results compared to typical discrete optimization algorithms. Because the motion of particles is
 307 continuous, the standard particle swarm encoding scheme cannot be used directly to solve ALBP-HRC.

Therefore, a method is needed to link the position of particles with their corresponding solutions. Among various methods, we have adopted the classical effective random key approach [47]. In subsection 4.3, the method for mapping the positions of particles into the encoding is described.

4.1. Encoding scheme

Proper encoding is crucial for effectively solving ALBP-HRC. Building upon the research of Li et al. [17], this study develops a new three-vector encoding and decoding scheme, enabling better correspondence to specific three-objective optimization ALBP-HRC and simpler mapping to real-world solutions. Figure 4 illustrates the encoding, where three vectors represented as resource vector (R), task sequence vector (T), and operation mode vector (O). The specific numerical meanings and allocation methods of each vector are elaborated in the subsequent paragraphs. While Figure 4 provides an overview of workstation locations for clarity, the precise allocation of workstations is determined during the decoding phase.

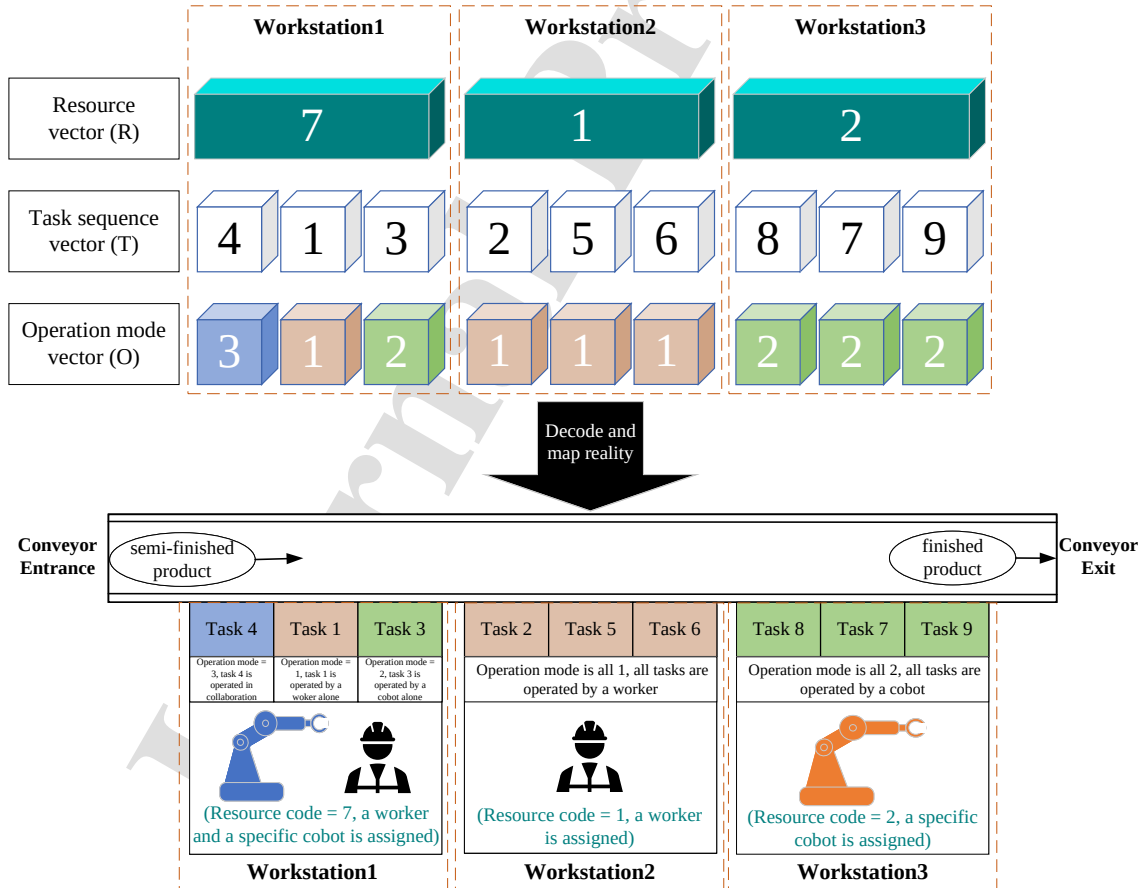


Figure 4: Encoding illustration

The first vector in the encoding is resource vector (R), which means workers and cobots are assigned to workstations as resources. Since there are NC types of cobots and one type of worker, there are $2NC+1$ resource alternatives available. Assuming a value in vector (R) is X , it can be one of the values in a set of resource schemes $\{1, 2, 3, \dots, 2NC+1\}$. The actual resource allocation corresponding to different X values is shown in Table 3.

Table 3: The description of a value of vector (R)

The value X in vector (R)	Description
$X=1$	A worker
$2 \leq X \leq NC + 1$	A specific type of cobot
$NC + 2 \leq X \leq 2NC + 1$	A combination of a worker and a specific type of cobot

The second vector in the encoding is task sequence vector (T), this indicates that tasks are placed in the workstation in a certain order, and executed sequentially. In the first workstation in Figure 4, task 4 is executed first, followed by task 1 and task 3. The assignment of task sequences needs to observe the constraint of precedence.

The third vector in the encoding is the operation mode vector (O). Each value in vector (O) can be either 1, 2 or 3. Assuming a value in vector (O) is Y , the operation mode of the task corresponding to different Y values can be found in Table 4. For example, in Figure 4, the first position in vector (T) and vector (O) are task 4 and operation mode 3 respectively. This indicates that task 4 is operated using operation mode 3, which corresponds to the human-robot collaboration mode. Similarly, because the value of vector (O) corresponding to task 3 is 2, it indicates that task is operated by a cobot alone. The primary significance of vector (O) is to determine the appropriate operation mode for a task located in a workstation.

4.2. Decoding scheme

Decoding scheme is aimed at transforming the encoding into an actual balancing scheme, while striving to achieve optimal values for each of the optimization objectives. One of the optimization objectives is the minimization of cycle time, and it can be regarded as an extension of the type-II assembly line balancing

Table 4: The description of a value of vector (O)

The value Y in vector (O)	Description
$Y=1$	The task is operated by a worker
$Y=2$	The task is operated by a cobot
$Y=3$	The task is operated collaboratively by a worker and a cobot

problem, which requires initial cycle time. For existing assembly lines, a fixed initial cycle time is available for optimization. However, for assembly lines that are yet to be designed, there is no predetermined initial cycle time, and therefore, a reasonable method is required to calculate the initial cycle time. A feasible scheme is to design a theoretical minimum initial cycle time CT_{tmint} , which gradually increases to an actual cycle time through iteration. CT_{tmint} is calculated using Equation 18.

$$CT_{tmint} = (\sum_{i=1}^{NT} \min_{1 \leq v \leq 2NC+1} t_{iv}) / NW \quad (18)$$

Through multiple experiments, this study found that for assembly lines with longer average operation times, the CT_{tmint} significantly differs from the actual cycle time. This discrepancy leads to excessive loop calculations during decoding and thereby reduces algorithm efficiency. Therefore, an initial cycle time CT_{int} update method based on the bisection method is designed to effectively reduce the number of loop calculations during decoding while maintaining good accuracy. The update of CT_{int} occurs after each iteration of the multi-objective optimization algorithm when a new Pareto front archive is obtained, as shown in Algorithm 1.

During the algorithm's iterations, violations of constraints may occur, meaning that the encoding does not directly map to a real-world assembly line balancing scheme. This study designed a corrective decoding scheme that allocates tasks and resources to workstations and determines the cycle time. The decoding procedure is shown in Algorithm 2.

Algorithm 1: Update scheme of initial cycle time

-
- Step 1:** Set $CT_{t_{mint}}$ based on Equation 18, set parameter $\alpha = 0$;
- Step 2:** Find the solution with the lowest CT from the g^{th} generation Pareto front archive, set this solution's CT as $CTlow_g$;
- Step 3:** If $CTlow_g < CTlow_{g-1}$, then $\alpha = \alpha - 1$ & $\alpha \geq 0$. Else if $CTlow_g = CTlow_{g-1}$, then $\alpha = \alpha + 1$;
- Step 4:** Update $CT_{int} = CTlow_g - (\frac{1}{2})^\alpha (CTlow_g - CT_{t_{mint}})$;
- Step 5:** If a new Pareto front archive is obtained during the iteration of the algorithm, repeat step 2;
-

Algorithm 2: Decoding procedure

-
- Step 1:** Set $CT_1 = CT_{int}$. Set workstation number $ws = 0$. Set a collection CS to hold the tasks that have no immediate predecessor tasks or all have all immediate predecessor tasks completed;
- Step 2:** If CS is null, go to step 6; otherwise, open a new workstation $ws = ws + 1$, set the workstation time $WT_{ws} = 0$;
- Step 3:** Select the task in CS that appears first according to the sequence in vector (T). If constraints allow, assign the task to the first position of workstation ws ; otherwise, adjust vector (R) and vector (O) to satisfy constraints, and then assign the task to the workstation ws ;
- Step 4:** Update the WT_{ws} . If $WT_{ws} > CT_1$, then cancel the previously assigned task and return to step 2;
- Step 5:** According to the sequence of tasks in vector (T), traverse the tasks in CS . If constraints allow, assign the task to workstation ws in order, then update CS and return to step 4;
- Step 6:** If all tasks in vector (T) are assigned, end the decoding procedure; otherwise, set $CT_1 = CT_1 + 1$, cancel all previous task allocations, $ws = 0$, initialize CS and return to step 2;
-

4.3. Particle initialization based on rule method and mapping

For the initial solution generation method, heuristic rules can typically provide a better initial solution. Based on the research of Rabbani et al. [48], this study employs ten heuristic task assignment rules in the initialization phase, which can significantly improve the diversity of the solutions. The ten rules are shown in Table 5.

Table 5: Ten task assignment rules

Rule number	Task assignment rules
1	Shortest Processing Time
2	Longest Processing Time
3	Minimum Total Number of Successor Tasks
4	Maximum Total Number of Successor Tasks
5	Maximum Total Time of Successor Tasks
6	Minimum Total Time of Successor Tasks
7	Maximum Total Number of Predecessor Tasks
8	Minimum Total Number of Predecessor Tasks
9	Maximum Total Time of Predecessor Tasks
10	Minimum Total Time of Predecessor Tasks

The Particle initialization procedure is presented in Algorithm 3. Here is an example in Figure 5 demonstrating how the initial position of a particle is generated, with the precedence relationship in this example being the same as in Figure 2. Additionally, the particle's position is solely related to vector (T), while the initial vectors (R) and (O) are randomly generated.

After the completion of the particle initialization phase, whenever the position of a particle changes, the sequence of codes in vector (T) is adjusted based on the values of particle's position vector and rearranged in descending order. For example, after a particle has been updated several times, the position vector changes to [0.752, 0.663, 0.615, 0.833, 0.811, 0.367, 0.446, 0.289]. Since the fourth number in the position vector is the largest number value (0.833 in position 4), the first value in the vector (T) changes to 4. Since the fifth number in the position vector is the second largest number (0.81 in position 4), the second value in the vector (T) changes to 5. Continuing this process sequentially, this vector (T) finally changes to [4,5,1,2,3,8,7,9].

Algorithm 3: Particle initialization procedure

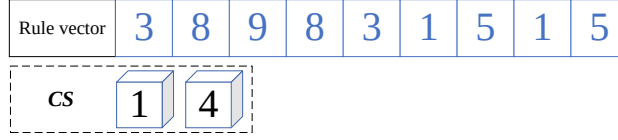
-
- Step 1:** Generate a rule vector of length NT , where each value in the vector is randomly generated integer within $[1,10]$. Traverse all unassigned tasks and add tasks to the candidate set (CS) if they have no immediate predecessors or if all their immediate predecessors have already been assigned;
- Step 2:** Suppose the current first value in the rule vector is FV . Select a task Ta from the CS using rule FV , and remove FV from the rule vector;
- Step 3:** Remove task Ta from CS , then add task Ta to vector (T);
- Step 4:** Update CS based on tasks that have not current immediate predecessors. If the rule vector is not null, return to step 2;
- Step 5:** If the rule vector is null and the length of vector (T) is NT , generate a position vector of length NT where each value is a random floating-point number between 1.000 and 0.001;
- Step 6:** According to the sequence of tasks in vector (T), adjust the positions of each value in the position vector so that tasks appearing earlier have larger corresponding floating-point numbers;
-

4.4. Local search scheme

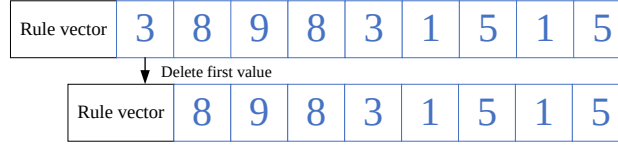
The previous sections explained that when the particle's position changes, its vector (T) will also be adjusted accordingly. The adjustments for vector (R) and vector (O) are performed during the local search phase, and this local search uses a heuristic rule to find high-quality feasible solutions.

Based on the bottleneck workstation and key workstation concepts in industrial engineering, adjusting the longest processing time workstation ($LTWS$) and shortest processing time workstation ($STWS$) is most likely to improve the balance rate of the assembly line. Since the processing time for each workstation has already been recorded as one of the solution attributes during step 4 of decoding, it is easy to find $LTWS$ and $STWS$. First, the local search attempts to make changes to the resources and operation mode codes from $LTWS$ and $STWS$ without violating any constraints. As adjustments are only made to $LTWS$ and $STWS$ for each solution without violating any constraints and each workstation has been assigned resources and tasks reasonably, there is no need for complex iterative decoding. Second, the partial particle's vectors (R) undergo mutation to enhance diversity. Finally, only recalculating objectives is required to obtain abundant effective neighbor solutions. After multiple experiments, this method has significantly increased efficiency

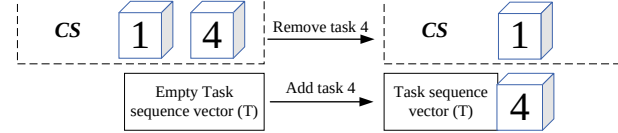
Step 1: Generate a rule vector and the task candidate set CS.



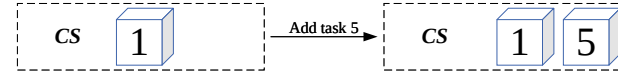
Step 2: Find $FV=3$. Based on Rule 3, since task 4 has fewer successor tasks than task 1 in CS, task 4 is selected. Remove 3 from rule vector.



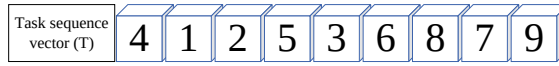
Step 3: Remove task 4 from CS, add task 4 to the vector (T).



Step 4: If rule vector is not empty, update CS.



Repeat steps 2 to 4, until the rule vector is empty. Then, obtain the final vector (T).



Step 5: Generate a position vector.



Step 6: Adjust the position vector through loops. Finally, the initial position of the particle is obtained.

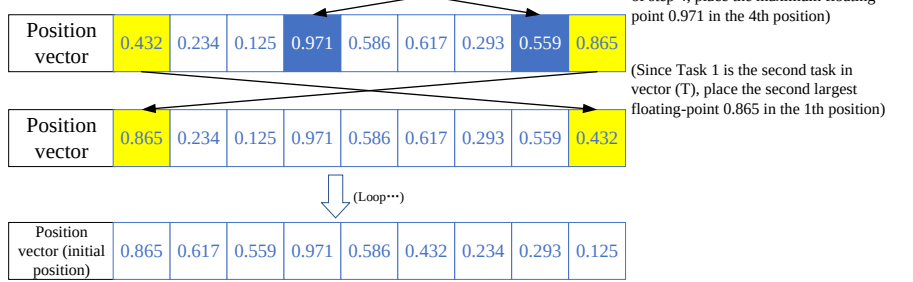


Figure 5: Steps 1 to 6 of a particle position initialization

388 compared to the traditional simple random local search. The local search procedure is shown in Algorithm

389 4.

Algorithm 4: Local search procedure**Input:** Population of particles (*POP*), Pareto front archive (*PFA*)**Output:** Pareto front archive (*PFA*)**begin**Set a empty new population (*NewPOP*);Get the size of *POP* as *ps*;**for** $i = 1 \rightarrow ps$ **do**Find the longest processing time workstation *LTWT*;Find the shortest processing time workstation *STWT*;**for** $j = 1 \rightarrow 2NC + 1$ **do****if** precedence constraint and operation constraint allow **then**Vector (R) in *LTWT* $\rightarrow j$;**if** the resource of the *LTWT* is a human and a cobot **then**| Switch the vector (O) in *LTWT*;**end if**Add the new solution into *NewPOP*;**end if****if** precedence constraint and operation constraint allow **then**Vector (R) in *STWT* $\rightarrow j$;**if** the resource of the *STWT* is a human and a cobot **then**| Switch the vector (O) in *STWT*;**end if**Add the new solution into *NewPOP*;**end if****end for****if** constraints allow **then**| Randomly change a resource in *POP_i*, then add it into *NewPOP*;**end if****end for**Get the size of *NewPOP* as *nps*;**for** $k = 1 \rightarrow nps$ **do**Calculate multi-objective values of *NewPOP_k*;Judge the Pareto between *NewPOP_k* and *PFA*;**end for****end**

5. Computational study

To validate the proposed method, this study selected instances from the well-known Scholl dataset (accessible at <https://assembly-line-balancing.de>) for testing. These instances offer explicit precedence relationships and worker operation times which are present in corresponding real-world instances, making them widely utilized in research on general assembly line balancing problems.

5.1. Experimental settings

Due to the complexity of the parameters in ALBP-HRC, some parameters necessitate random generation. In general, cobots have longer operation times compared to human workers [16], the study set that a cobot requires 1.3 to 1.6 times operation time than an individual worker to complete a task independently [6]. As human-robot collaboration combines the advantages of both humans and cobots [16], tasks operated in this mode typically require only 60% to 80% of the time needed for individual human operations. However, it is important to note that cobots may face limitations when independently processing certain tasks, and likewise, not all tasks are suitable for human-robot collaboration. As such, approximately 28% of randomly selected tasks cannot be exclusively operated by cobots, and around 21% of tasks cannot be operated in human-robot collaboration mode. The operation time will be assigned an infeasible value if the task cannot be operated by a specific operation mode. The energy consumption of workers independently processing tasks is generated using a uniform distribution $u(0.1, 2.0)$. Compared to workers independently processing tasks, using a human-robot collaboration model can reduce energy consumption by 30% to 80% [16]. The experiment adjusts cobot parameter settings based on the research by Li et al. [41]. There are not exceeding 4 types of available cobots, each with distinct annual comprehensive costs and capabilities. The annual comprehensive costs of the cobots are generated randomly, $AC_c = 4 + 2 \times c \times Rand[0.8, 1.2]$, $AS = 8$. The annual comprehensive cost and capability of the cobots are positively correlated, meaning that more expensive cobots tend to have shorter operation times [41], and the operation time should be controlled within the previously mentioned range. In the human-robot collaboration mode, the worker's energy consumption is positively correlated with operation time, and considering potential variations in actions, a fluctuation of approximately 25% is allowed.

Since the three-objective optimization problem in this study is special, comparing the proposed IMOPSO with classical multi-objective optimization algorithms [49] in authoritative papers, including MABC [50], MOSA [51] and NSGA-II [52], can effectively demonstrates the effectiveness of IMOPSO. As a frontier algorithm, POGE [53] is also used as a comparison algorithm to prove the advanced quality of IMOPSO. All algorithms are implemented and executed in C++ programming language. The computer used for the validation experiments has a CPU with a clock speed of 3.5 GHz and 32 GB RAM. Visual Studio 2022 is used as the IDE tool for all algorithms. Based on the problem characteristics, the termination criterion for each algorithm is set such that the computation time does not exceed $NT \times NT \times \theta$ ms [54]. NT is the number of workstations in the instance, $\theta \in \{10, 20, 30\}$.

5.2. Evaluation metrics

Compared to single-objective optimization problems, multi-objective optimization problems require the use of specific metrics to evaluate the quality of non-dominated solution sets. Therefore, this study selects three commonly used and recognized indicators: hypervolume ratio (HVR) [55], Spacing [56] and inverted generational distance (IGD) [57]. These three metrics, which can cover the entire Pareto front and accurately reflect the distribution of the algorithm in the search space, have been widely used in multi-objective optimization problems of assembly line balancing [6, 58]. HVR quantifies the quality of the obtained Pareto front by calculating the ratio between the hypervolume of the obtained Pareto front and the hypervolume of the real Pareto front. A higher value of HVR indicates better performance of multi-objective optimization algorithms. HVR is defined in Equation 19.

$$HVR = \frac{volume(\bigcup_{m=1}^{|P|} v_m)}{volume(\bigcup_{n=1}^{|R|} v_n)} \quad (19)$$

Where, *volume* means the calculation of the volume size. P is the obtained Pareto front, R is the real Pareto front, $|P|$ is the number of the solutions in P , $|R|$ is the number of the solutions in R , v_m and v_n are the hypercube formed by the m -th and n -th solution and the reference point respectively.

The Spacing metric is a value reflecting the distribution of vectors throughout the obtained Pareto front. The Pareto front solution set with a uniform distribution can provide more diversified choices and better reflect the trade-off relationship among multiple objective functions. The calculation of Spacing is based on

the summation of distances between each solution in the set and its nearest neighbor. Smaller Spacing values are better. The Spacing calculation method is defined by Equation 20.

$$Spacing = \sqrt{\frac{1}{|P| - 1} \sum_u^{|P|} (\bar{d} - d_u)^2} \quad (20)$$

Where, $|P|$ is the number of the obtained Pareto front P , d_u represents the minimum distance from the u -th solution in P to other solutions, $\bar{d}$ is the mean value of all d_u .

IGD is a commonly used performance metric in multi-objective optimization. IGD evaluates the diversity and convergence of multi-objective optimization algorithms by calculating the distance between the real Pareto front and the generated solution set. Smaller IGD values indicate a better algorithm. The IGD calculation method is defined by Equation 21.

$$IGD(P, R) = \frac{1}{|R|} \sum_{m=1}^{|R|} \min_{n \in P} d(m, n) \quad (21)$$

Where, P is the obtained Pareto front, R is the real Pareto front, $|R|$ is the number of the solutions in R , $d(m, n)$ is the distance between the m -th solution and the n -th solution.

5.3. Parameter tuning

In order to ensure the high performance of the algorithm, this study utilizes the well-known Taguchi method [59] for parameter calibration. The Taguchi method conducts experiments by setting reasonable parameter values combinations to obtain a measurement called signal-to-noise (S/N) ratio. This study selected three cases of varying scales to evaluate the parameters, including a small-scale case (Jackson) with 11 tasks, a medium-scale case (Warnecke) with 58 tasks, and a large-scale case (Bartholdi1) with 148 tasks. Each case undergoes 10 tests, and the average HVR of all results is calculated. Considering that a higher HVR value indicates a better performance, Equation 22 is employed to calculate the S/N ratio.

$$S/Nratio = -10 \times \lg(1/\overline{HVR})^2 \quad (22)$$

For IMOPSO, the four parameters that mainly impact performance are population size (PS), inertia factor ω , learning factors c_1 and c_2 . Referring to the existing research on MOPSO algorithms and after brief testing, this study set four levels for each parameter, $PS \in \{20, 40, 60, 80\}$, $\omega \in \{0.6, 0.8, 1.0, 1.2\}$,

$c_1 \in \{0.5, 1.0, 1.5, 2.0\}$, $c_2 \in \{0.5, 1.0, 1.5, 2.0\}$. The orthogonal array table of the Taguchi experiment is shown in Table 6. The termination criterion for each algorithm is set to not exceeding $NT \times NT \times 30$ ms.

Table 6: The orthogonal array

Test number	PS	ω	c_1	c_2
1	20	0.6	0.5	0.5
2	20	0.8	1.0	1.0
3	20	1.0	1.5	1.5
4	20	1.2	2.0	2.0
5	40	0.6	1.0	1.5
6	40	0.8	0.5	2.0
7	40	1.0	2.0	0.5
8	40	1.2	1.5	1.0
9	60	0.6	1.5	2.0
10	60	0.8	2.0	1.5
11	60	1.0	0.5	1.0
12	60	1.2	1.0	0.5
13	80	0.6	2.0	1.0
14	80	0.8	1.5	0.5
15	80	1.0	1.0	2.0
16	80	1.2	0.5	1.5

The results of the Taguchi test are shown in Figure 6. Based on the results, the population size of IMOPSO significantly impacts on the stability and performance of the algorithm. The stability of the algorithm increases with a higher S/N value. Equation 22 shows that the average HVR in orthogonal experiments is directly proportional to the S/N ratio. Therefore, it is advisable to select parameters with higher S/N values,

470 $PS = 40$, $\omega = 1.0$, $c_1 = 1.5$, $c_2 = 1.0$. Finally, using a similar approach, the parameters for all algorithms
 471 were obtained as shown in Table 7.

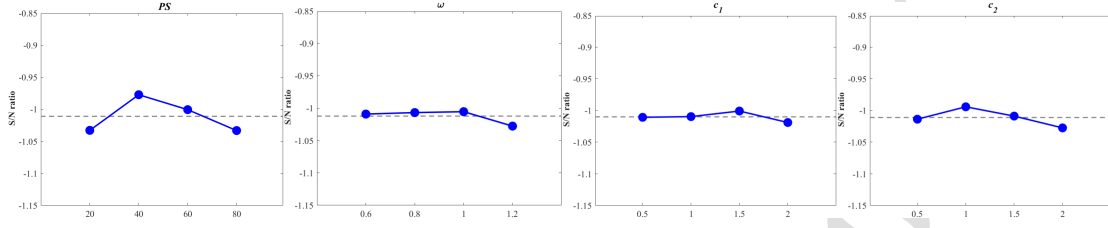


Figure 6: The mean S/N ratio plot of Taguchi test

Table 7: Parameters table

	IMOPSO	MABC	MOSA	NSGA-II	POGE
Parameter setting	$PS = 40$	$PS_{employed} = 40$	$T_0 = 100$	$PS = 80$	$PS = 80$
	$\omega = 1.0$	$PS_{onlookers} = 30$	$T_f = 10$	Crossover rate = 0.8	
	$c_1 = 1.5$	Crossover rate = 0.5	Cool rate = 0.9	Mutation rate = 0.2	
	$c_2 = 1.0$				

472 5.4. Effectiveness of the IMOPSO algorithm

473 The experiments utilized 9 precedence diagrams from the Scholl dataset, with the NT and NW of each
 474 instance determined. These 9 instances can be categorized into three different scales based on the number of
 475 tasks: small-scale instances ($NT < 20$), medium-scale instances ($20 \leq NT \leq 80$), and large-scale instances
 476 ($80 < NT$). Each scale contains 3 instances. Each algorithm is executed 10 times to obtain the average
 477 value. Table 8, Table 9 and Table 10 display the results of three metrics for five under various θ .

478 Based on Figure 7, Figure 8, and Figure 9, IMOPSO outperforms the other algorithms in 71 out of
 479 81 comparisons across the HVR, Spacing, and IGD metrics, accounting for 87.7%, thereby validating the
 480 effectiveness of IMOPSO. When $\theta = 20$ or 30, IMOPSO exhibits significantly higher HVR values compared
 481 to conditions at $\theta = 10$, indicating that IMOPSO avoids premature convergence and maintains strong search
 482 capabilities over longer computational times. IMOPSO demonstrates superior performance in moderate and
 483 large-scale instances compared to small-scale instances, suggesting it is more suitable for handling larger-
 484 scale problem. Additionally, it is noteworthy that the performance of the other four algorithms displays
 485 considerable variability across different instances, possibly due to changes in instance characteristics. In

contrast, IMOPSO consistently delivers good results, demonstrating its excellent stability and generality.

To more intuitively illustrate the efficiency and accuracy of the algorithms, Figure 10 presents the results under different termination criteria determined by parameter θ . The vertical axis represents the mean evaluation metric across all instances. According to the termination criterion, the computation time is positively correlated with θ . In contrast to single-objective optimization, complex multi-objective optimization algorithms generate Pareto fronts that contain many non-dominated solutions. As a result, achieving ideal convergence within a limited time is challenging. However, the efficiency of the algorithm can still be evaluated by comparing the results obtained at different computation times. From Figure 10, it can be observed that as the computation time increases, the HVR values for all five algorithms show an increasing trend, while the IGD values exhibit a decreasing trend. Regarding the mean value of the IGD metric, the results of IMOPSO with a shorter computation time ($\theta = 10$) outperform the comparison algorithms that require more computation time, thus demonstrating that the IMOPSO algorithm is more computationally efficient. Under the same θ , the HVR and IGD values for IMOPSO are significantly better than those of the comparison algorithms. Since HVR and IGD are mainstream quality assessment metrics for multi-objective optimization, this suggests that IMOPSO also performs well in terms of result accuracy. Additionally, IMOPSO consistently achieves lower Spacing values across all three θ settings, demonstrating the good diversity of the Pareto solutions. In summary, IMOPSO is more capable of generating diverse and high-quality non-dominated solutions within a limited time, making it more suitable for practical applications.

5.5. Relationship analysis of multiple objectives

Analyzing the trade-off relationships among multiple objectives deepens our understanding of the problem. In this study, the correlation coefficient matrix [60] is employed to demonstrate the correlational relationships among the three objectives. Each element of the matrix represents the correlation between the objective variables indicated by the rows and columns. If the correlation between two variables is close to -1, then there exists a strong negative correlational relationship between them, that is to say, when one variable increases, the other variable will decrease. If the correlation between two variables is close to 1, then there exists a strong positive correlational relationship between them. If the correlation is close to 0, there is almost no linear relationship between these two variables.

Table 8: Results of three metrics for five algorithms when $\theta = 10$

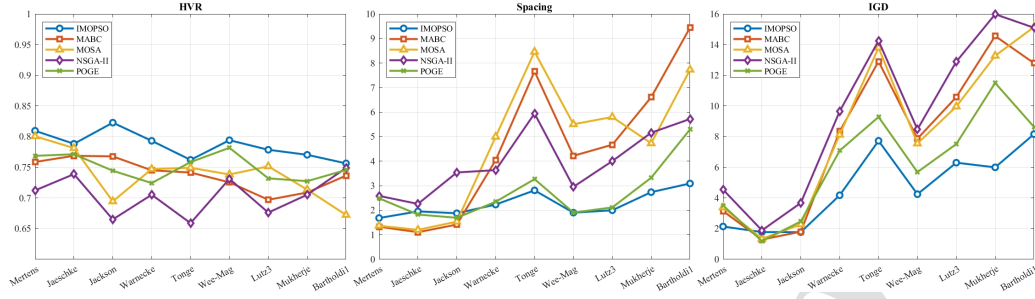
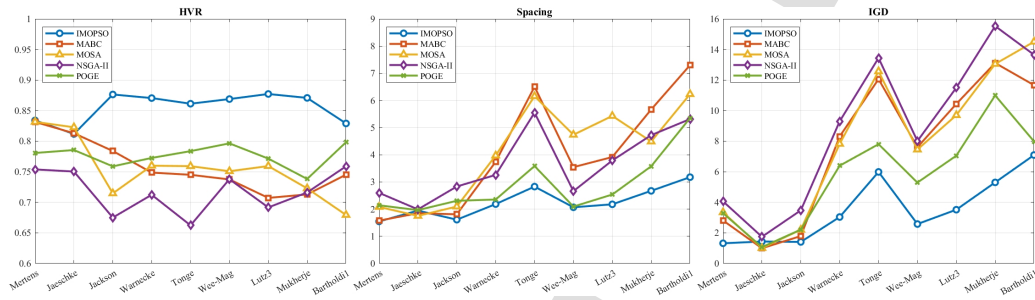
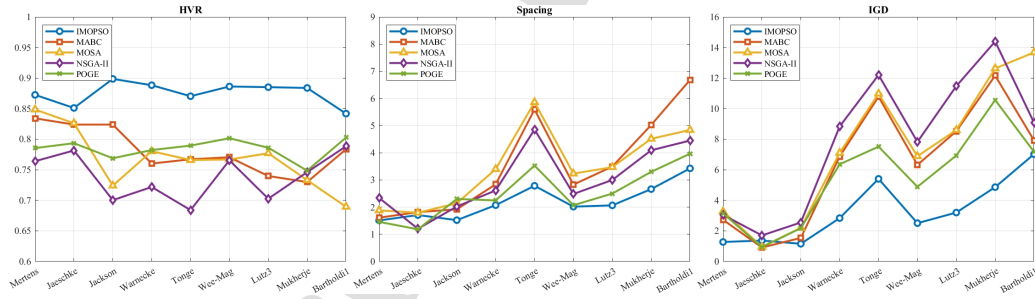
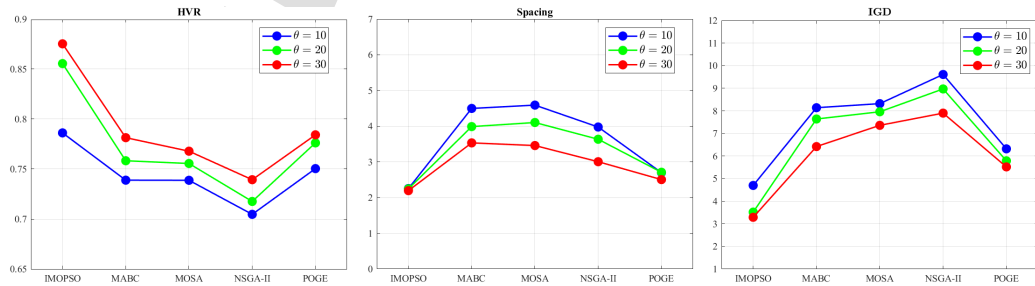
instance	NT	NW	HVR	Spacing					IGD				
				IMOPSO	MABC	MOSA	NSGA-II	POGE	IMOPSO	MABC	MOSA	NSGA-II	POGE
Mertens	7	5	0.8096	0.7587	0.8006	0.7123	0.7687	1.6828	1.3286	1.3684	2.5855	2.4869	2.1396
Jaeschke	9	4	0.7883	0.7688	0.7814	0.7389	0.7713	1.9526	1.1054	1.2009	2.2583	1.8246	1.7771
Jackson	11	8	0.8228	0.7678	0.6945	0.665	0.7444	1.8769	1.4151	1.5362	3.5394	1.6924	1.7728
Warnecke	58	17	0.7932	0.7451	0.7471	0.7053	0.7243	2.232	4.0406	4.9952	3.6356	2.3456	4.1774
Tonge	70	21	0.7621	0.7415	0.7493	0.659	0.7585	2.8105	7.6592	8.4584	5.9433	3.268	7.7292
Wee-Mag	75	15	0.7942	0.7255	0.7383	0.7312	0.7818	1.9009	4.2187	5.5053	2.9538	1.9048	4.2463
Lutz3	89	21	0.7786	0.6971	0.7516	0.6762	0.732	1.9985	4.6699	5.7996	4.0033	2.1151	6.3016
Mukherje	94	16	0.7704	0.7089	0.7136	0.7053	0.7274	2.7374	6.6045	4.7283	5.1591	3.3247	5.997
Bartholdi1	148	10	0.7564	0.7364	0.6724	0.7489	0.7456	3.0912	9.4424	7.7291	5.7195	5.2943	8.1642

Table 9: Results of three metrics for five algorithms when $\theta = 20$

instance	NT	NW	HVR	Spacing					IGD				
				IMOPSO	MABC	MOSA	NSGA-II	POGE	IMOPSO	MABC	MOSA	NSGA-II	POGE
Mertens	7	5	0.834	0.8315	0.8316	0.7538	0.7807	1.5462	1.5756	2.0795	2.598	2.1398	1.3178
Jaeschke	9	4	0.8118	0.813	0.8227	0.7503	0.7855	1.944	1.8484	1.7502	1.989	1.9625	1.4337
Jackson	11	8	0.8764	0.7839	0.7146	0.675	0.7588	1.6105	1.8078	2.0939	2.8318	2.3056	1.4071
Warnecke	58	17	0.8705	0.7486	0.7599	0.7123	0.7726	2.1843	3.7409	3.9711	3.2583	2.3525	3.0372
Tonge	70	21	0.8613	0.745	0.7591	0.6623	0.7837	2.8281	6.5058	6.1554	5.5513	3.5838	5.9887
Wee-Mag	75	15	0.8689	0.7374	0.7508	0.7378	0.7964	2.0665	3.5439	4.7354	2.6607	2.0926	2.5772
Lutz3	89	21	0.8772	0.7072	0.7593	0.6916	0.7715	2.176	3.9159	5.4288	3.7905	2.5362	3.5161
Mukherje	94	16	0.8708	0.7129	0.7228	0.7161	0.7382	2.6715	5.666	4.4877	4.7215	3.5754	5.3031
Bartholdi1	148	10	0.829	0.7452	0.6793	0.7589	0.7983	3.1756	7.2991	6.2312	5.3134	5.3517	7.0927

Table 10: Results of three metrics for five algorithms when $\theta = 30$

instance	NT	NW	HVR	Spacing					IGD				
				IMOPSO	MABC	MOSA	NSGA-II	POGE	IMOPSO	MABC	MOSA	NSGA-II	POGE
Mertens	7	5	0.8725	0.8341	0.8485	0.764	0.7854	1.5083	1.8787	1.6104	2.338	1.4558	1.2726
Jaeschke	9	4	0.8511	0.8239	0.8261	0.7815	0.7934	1.7081	1.8141	1.7933	1.202	1.1836	1.3669
Jackson	11	8	0.8986	0.8239	0.724	0.7003	0.7688	1.519	1.9193	2.1398	2.0211	2.2989	1.1543
Warnecke	58	17	0.8883	0.7601	0.7803	0.7218	0.7822	2.0708	2.8378	3.3983	2.6128	2.246	2.8351
Tonge	70	21	0.8703	0.7671	0.7659	0.684	0.7895	2.7835	5.5929	5.8611	4.8559	3.5277	5.4031
Wee-Mag	75	15	0.8862	0.7704	0.7667	0.7654	0.8015	2.0158	2.8268	3.2325	2.4837	2.0683	2.5043
Lutz3	89	21	0.8851	0.7397	0.777	0.7026	0.786	2.0641	3.4921	3.4701	2.997	2.4976	3.1981
Mukherje	94	16	0.8838	0.7299	0.7329	0.7461	0.7485	2.6625	5.0222	4.5147	4.0916	3.297	4.862
Bartholdi1	148	10	0.8418	0.7835	0.6894	0.7884	0.8027	3.4251	6.6822	4.8376	4.4488	3.9646	7.0195

Figure 7: Line graph of 5 algorithms for 9 instances when $\theta = 10$ Figure 8: Line graph of 5 algorithms for 9 instances when $\theta = 20$ Figure 9: Line graph of 5 algorithms for 9 instances when $\theta = 30$ Figure 10: The mean results for 5 algorithms under three termination criteria with different θ

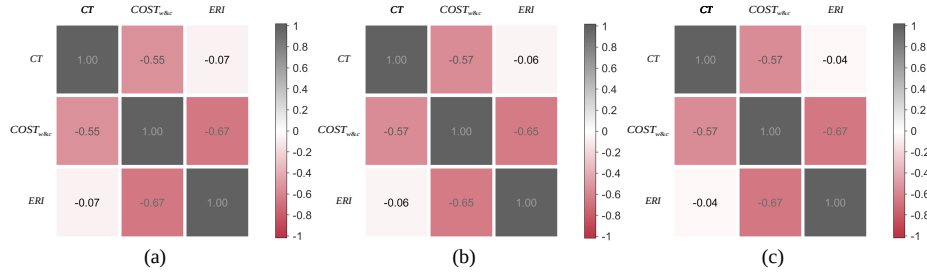


Figure 11: The correlation coefficient matrix between the three objectives

The results of the correlation experiments show that there is a strong negative correlation between production cycle time and cost. This is because more resources lead to an improvement in production efficiency, which is consistent with normal production situations. There is also a strong negative correlation between cost and ergonomic risk. This is due to the fact that the increase in cost raises the level of automation and reduces the workload of workers. Interestingly, there is a weak correlation between production cycle time and ergonomic risk. This may be because of the mutual influence of the two situations. On the one hand, if cobots are used to completely replace workers' labor as much as possible, the ergonomic risk will be significantly reduced, but the efficiency of cobots operating tasks independently is not high. On the other hand, if workers are allowed to operate tasks as much as possible, the ergonomic risk will be the highest, but the production efficiency will be slightly higher than that in the previous situation. The results of the experiments on medium-sized instances are shown in Figure 11. In addition, this study expanded on the aspects that have not been discussed in previous studies, that is, the impact of the operation constraint characteristics of the assembly line instance on the multi-objective trade-off. Figure 11 (a) shows the result of reducing the proportion of tasks that cannot be exclusively operated by cobots and the proportion of tasks that cannot be operated by human-robot collaboration mode by about 10% each. (b) shows the result of maintaining the two proportions at about 28% and 21% respectively. (c) shows the result of increasing each of the two proportions by 10%. As the proportion of tasks subject to the operation constraint increases, there is a slight tendency for the correlation between cycle time and ergonomic risk to weaken. This perhaps indicates that ALBP-HRC requires more real-world cases to expand its research value. But overall, the experimental results show that the change in the number of operationally constrained tasks does not significantly affect

the correlation coefficient between the objectives.

6. Case application

A study on an existing electric vehicle air conditioner assembly line is conducted in this section. The data for electric vehicle air conditioners were obtained through field investigation. Considering the availability of three types of cobots for procurement, this section designed multiple human-robot collaborative assembly line balancing schemes for the factory for evaluation.

6.1. Information of the case

The market for electric vehicles is expanding due to their advantages in reducing air pollution and alleviating energy shortages [61]. Therefore, this electric vehicle air conditioner assembly line, operated entirely by workers, has encountered issues of insufficient capacity and high labor load. Due to space constraints in the factory, expanding the number of workstations freely is not feasible. Furthermore, the inherent characteristics of air conditioner assembly tasks make full automation impractical for the automotive air conditioner assembly line. Compared to the conventional car air conditioners, the M-type electric vehicle air conditioner has more precision components and is subject to more constraints. The exploded view of the electric vehicle air conditioner is shown in Figure 12, which includes 31 major components. For clarity, please note that sponges, screws, and some micro sensors are not depicted. Some plastic components, such as parts No. 8 and No. 20, are prone to slight deformations like bending due to their inherent manufacturing process and logistics transportation characteristics. The larger the plastic component, the more noticeable the deformation. Because workers' assembly actions are highly flexible, they can easily and correctly assemble these slightly deformed components. However, when cobots assemble slightly deformed plastic parts, two scenarios can occur. In the first scenario, the plastic part is considered defective and not assembled. In the second scenario, the plastic part is considered acceptable, but the assembly action cannot align accurately. For the assembly of certain parts, such as part No. 28, it is very challenging for cobots to tighten screws inside the air conditioner box. This task is also not suitable for a human-robot collaboration mode and is better suited for individual worker operations. Some components can be effectively used in human-robot collaboration scenarios, such as combining parts No. 26 and No. 27. This assembly task requires flipping the air conditioner

box, and utilizing the help of a cobot in this process can significantly improve efficiency. In fact, this factory previously experienced a failed fully automated assembly line, mainly because the robots lacked flexibility in executing certain tasks, which reduced assembly quality, and the actual production efficiency did not meet expectations. In summary, introducing cobots at a reasonable cost and designing a new balancing scheme for the electric vehicle air conditioner assembly line is an applicable method.

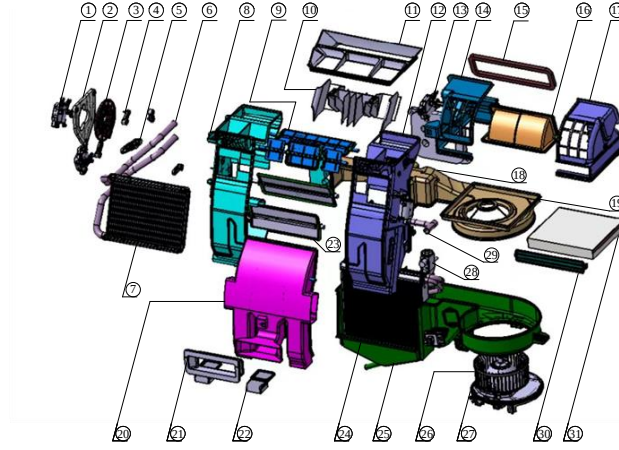


Figure 12: The exploded view of the electric vehicle air conditioner

6.2. Result of case application

There are a total of 46 assembly tasks on the electric vehicle air conditioner assembly line with 11 workstations, and information on operation constraint and standard operation time is surveyed. The precedence relationship diagram of the electric air conditioner is shown as Figure 13, Appendix A provide the operation times. Tasks marked in yellow on the precedence relationship diagram cannot be operated by cobots alone, which often involve assembling plastic parts with high variability. The tasks marked in gray cannot be operated using human-robot collaboration mode, as these tasks usually involve assembling and processing delicate small parts, which may result in collisions between humans and cobots. Tasks marked in red can only be operated by workers individually, requiring a high level of flexibility. Three types of cobots are available: cobot1, cobot2, and cobot3, with annual comprehensive costs of 6, 7.5, and 9 respectively, and higher costs correspond to greater capabilities, $AS = 8$. Other parameters can be referenced from the settings in subsection 5.1. Each algorithm is set with a termination criterion of $NT \times NT \times 30$ ms. Figure 14 Fig.

14 presents the Pareto front obtained from a single run of five algorithms. It can be observed that, in the three-dimensional perspective, IMOPSO is positioned closer to the lower left corner, indicating that it is more likely to obtain solutions near the real Pareto front. Additionally, IMOPSO generates more solutions within a fixed time and tends to produce solutions closer to either the left or bottom axis, making it more capable of obtaining distinctive solutions. Figure 15 presents the boxplots of the three metrics across 20 runs of the 5 algorithms. It is evident that IMOPSO outperforms the other algorithms across all three metrics, indicating that IMOPSO is suited for solving this case and finding high-quality schemes.

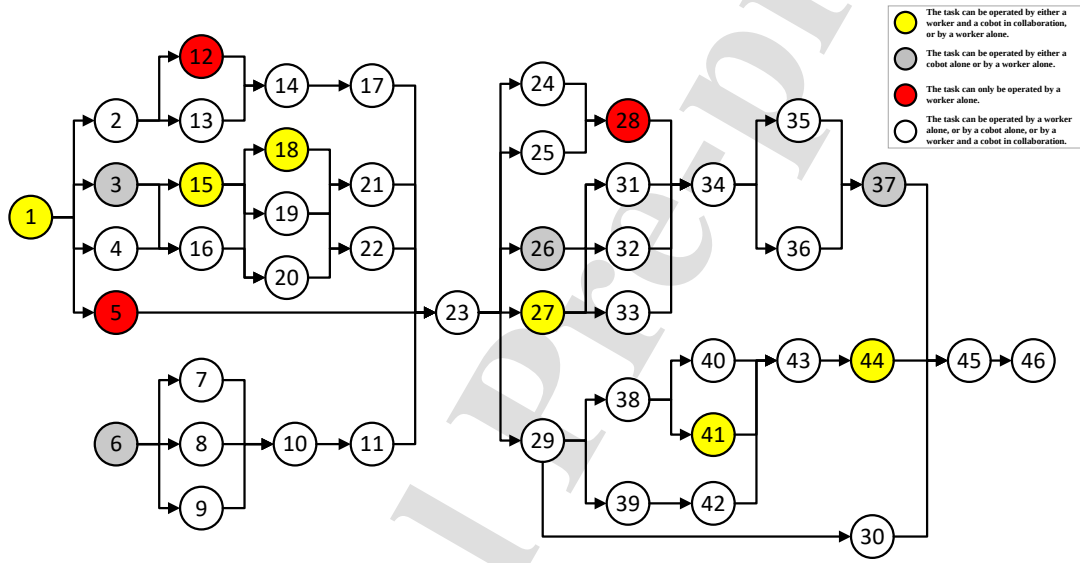


Figure 13: The precedence relationship diagram of the electric vehicle air conditioner

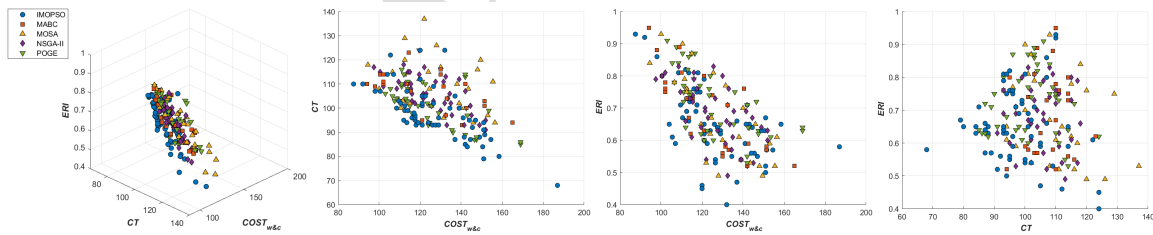


Figure 14: Pareto front of five algorithms run once in the case application

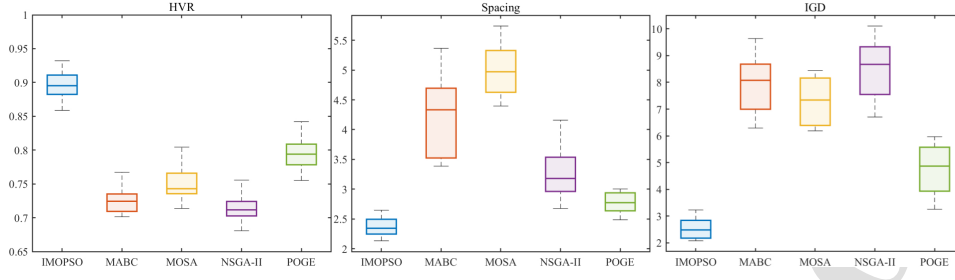


Figure 15: Boxplots of the three metrics for five algorithms run 20 times

6.3. Analysis of assembly balancing schemes

Before introducing cobots and balancing, the initial data for this M-type electric vehicle air conditioner assembly line were as follows: $CT = 124$, $COST_{w\&c} = 88$, $ERI = 1$. By introducing cobots and utilizing the IMOPSO, a Pareto front containing multiple balancing schemes is obtained. Although different schemes are non-dominated solutions, based on the idea of the relative percentage deviation, one can use the average of the ratios of the three objectives of a scheme to the original objective values to obtain an evaluation index EI . The EI measures whether the solution is relatively excellent in all three objectives, and the calculation method is Equation 23.

$$EI = \sum_{n=1}^{|N|} \frac{Ob_{some}^n}{Ob_{initial}^n} \quad (23)$$

Where, $|N|$ is the number of optimization objectives. Ob_{some}^n is the value of the n-th optimization objective in a solution. $Ob_{initial}^n$ is the value of the n-th optimization objective in initial data.

Based on EI values, two representative schemes are selected from the Pareto front set. In the scheme diagrams, brown indicates tasks operated solely by workers, green indicates tasks operated solely by robots, and blue indicates tasks operated in a human-robot collaborative mode. Scheme A ($EI = 2.73$), as shown in Figure 16, focuses on reducing ergonomic risk. Its ERI decreased by 54%, $COST_{w\&c}$ increased by 36%, and CT decreased by 10%. Scheme B ($EI = 2.80$), as shown in Figure 17, focuses on reducing cycle time. Its CT decreased by 24%, $COST_{w\&c}$ increased by 41%, and ERI decreased by 37%. The factory can select the appropriate scheme based on different requirements.

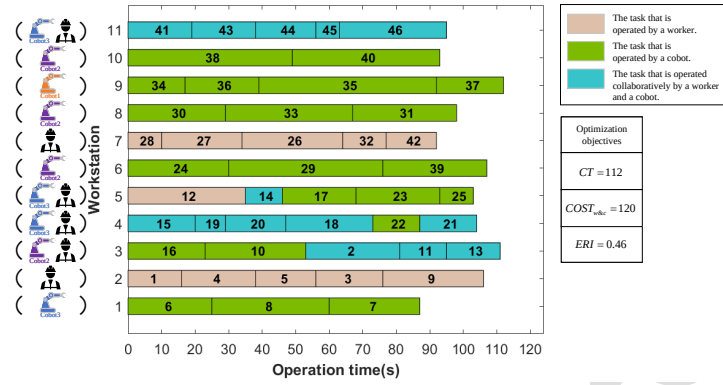


Figure 16: Trade-off scheme A that focuses on reducing ergonomic risk

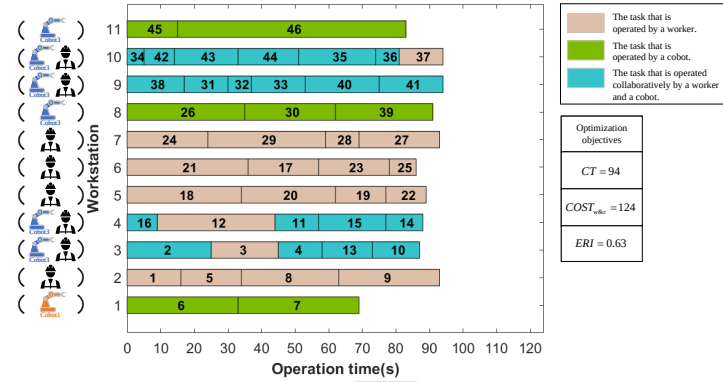


Figure 17: Trade-off scheme B that focuses on reducing cycle time

6.4. Application Discussion

The case application supports the feasibility of the proposed method in assisting SMEs with practical assembly production. In most SMEs, high-workload manual assembly lines remain the dominant assembly production mode. SMEs are generally reluctant to rashly invest in a large number of expensive robots. Fully automated production is also difficult to adapt to SMEs due to its lack of flexibility and real-time responsiveness. However, the technological advancement in human-robot collaboration brought by Industry 5.0 is expected to improve the existing production models. The proposed Model of ALBP-HRC and IMOPSO algorithm can help SMEs obtain effective human-robot collaboration balancing schemes. These balancing schemes enable the introduction of a limited number of cobots at an appropriate cost, achieving the goal of improving production efficiency and reducing human-related risks while retaining the flexibility of the assembly line. Computational experiments further demonstrate that the IMOPSO algorithm is more competitive

in terms of both computation time and solution quality. With the increasing adoption of cobots, this study can make a valuable contribution to building more efficient and human-centered assembly lines for SMEs.

7. Conclusion and future research

The ALBP-HRC is a complex combinatorial optimization problem. The study made an investigation of the assembly line of an electric vehicle air conditioner factory. Based on the real requirements of the factory, the study developed a novel ALBP-HRC mathematical model with the objective functions of minimizing cycle time, ergonomic risk and cost. To address the characteristics of the ALBP-HRC, a novel encoding and decoding method is proposed, and an effective IMOPSO algorithm is designed. Experiments show that the proposed model and the solution approach are superior to the existing similar algorithms in finding high-quality schemes within a limited time. This study focuses on typical assembly line production in SMEs, where the tasks are not complex, and therefore the skill differences among workers are not specifically considered, which is a limitation of the study. Due to the differences between SMEs and large enterprises, the ALBP-HRC in large enterprises involves longer operation times. The existing model can be further extended to consider factors such as worker learning curves, potential disruptions, or parallel task completion in specific conditions, which has the potential to advance ALBP-HRC research in different production scenarios.

The research on ALBP-HRC is still in its early stage. To address some more realistic scenarios, the fuzzy operation times of tasks caused by the degree of human-robot collaboration could be considered. More on-site investigations and practices are needed for the research in this aspect. Furthermore, the shape and mode of the assembly line may also have an impact on the optimization scheme. Since a lot of research are based on the straight-line assembly lines with human-robot collaboration, there are various types of variants in the actual production environment. For example, the U-shaped assembly lines existing in medium-sized enterprises are also suitable for adapting to the human-robot collaboration mode. For a real and complex U-shaped assembly line scenario, how to successfully introduce the human-robot collaboration mode and effectively achieve the balance optimization goals is an interesting and challenging research topic.

Appendix A. Operation times

The Appendix provides the operation times of 46 tasks under 7 resource configurations.

Task	Operation times of tasks in different resources (s)						
	worker	cobot1	cobot2	cobot3	worker& cobot1	worker& cobot2	worker& cobot3
1	16	-	-	-	12	10	9
2	38	56	50	45	32	28	25
3	20	29	26	23	-	-	-
4	22	34	30	27	17	15	13
5	18	-	-	-	-	-	-
6	25	33	31	25	-	-	-
7	23	36	31	27	17	15	12
8	29	45	40	35	20	16	13
9	30	48	41	37	22	20	18
10	23	34	30	27	16	15	14
11	24	39	34	30	16	14	13
12	35	-	-	-	-	-	-
13	30	47	41	38	20	16	15
14	19	30	27	23	16	14	11
15	38	-	-	-	28	26	20
16	16	24	23	21	13	10	9
17	21	29	27	22	15	11	9
18	34	-	-	-	28	27	26
19	15	23	20	19	11	10	9
20	28	41	35	32	23	20	18
21	36	53	48	41	28	22	17
22	12	17	16	14	10	8	7
23	21	30	27	25	17	15	12
24	24	33	30	28	19	17	14
25	8	14	13	10	6	5	4
26	30	48	41	35	-	-	-
27	24	-	-	-	17	16	12
28	10	-	-	-	-	-	-
29	35	52	46	41	24	18	16
30	22	31	29	27	17	16	15
31	22	36	31	29	17	16	13
32	13	18	16	15	10	8	7
33	27	45	38	34	21	18	16
34	11	17	15	13	8	6	5
35	35	53	48	41	28	27	23
36	13	22	19	17	10	8	7
37	13	20	18	16	-	-	-
38	33	53	49	42	22	20	17
39	24	34	31	29	17	13	11
40	30	49	44	39	25	24	22
41	31	-	-	-	22	21	19
42	15	23	22	19	12	11	9
43	37	58	53	48	28	24	19
44	34	-	-	-	28	23	18
45	12	19	17	15	9	8	7
46	58	88	80	68	41	36	32

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Research on the balancing problem of human-robot collaborative assembly line in SMEs considering ergonomic risk and cost

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Highlights

Research highlight 1: A novel multi-objective human-robot collaborative assembly line problem for small and medium-sized enterprises is proposed.

Research highlight 2: A novel three-vector encoding and decoding scheme and an improved multi-objective particle swarm optimization algorithm are proposed to solve the problem.

Research highlight 3: The performance of the proposed algorithm is demonstrated through computational experiments and a real-world electric vehicle air conditioner assembly line case.